

RCIM Model-Bank Reproduction Forward Last Non-Green Cells Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: track1_forward_last_non_green_cells_campaign_2026-04-28_11_36_05
- planning report: doc/reports/campaign_plans/track_1/exact_paper/2026-04-28-11-21-25_track1_forward_last_non_green_cells_campaign_plan_report.md
- queue size: 108
- execution window: 2026-04-28T11:43:59.1021956+02:00 to 2026-04-28T14:08:27+02:00
- campaign output directory: output/training_campaigns/track1/exact_paper/forward_last_non_green_cells/track1_forward_last_non_green_cells_campaign_2026-04-28_11_36_05
- validation root: output/validation_checks/paper_reimplementation_rcim_original_dataset_exact_model_bank_forward_last_non_green_cells
- report timestamp: 2026-04-28-16-12-00

Executive Summary

- The final forward-only last non-green wave completed the full 108/108 queue successfully.
- Targeted family-target pairs: 7 .
- Promoted pair winners: 3 .
- Retained baseline pair winners: 4 .
- Forward non-green cells moved from 10 to 9 .

Family Best Retry Outcome

Family	BestRun	Scope	Harmonic	ClosureScore	Met	Near	Open
ERT	track1_forward_ert_ampl_h162_last_non_green_cells_attempt_15	amplitude	162	0.750	1	1	0
GBM	track1_forward_gbm_ampl_h162_last_non_green_cells_attempt_15	amplitude	162	0.000	0	0	2
XGBM	track1_forward_xgbm_ampl_h240_last_non_green_cells_attempt_08	amplitude	240	0.750	1	1	0
LGBM	track1_forward_lgbm_ampl_h0_last_non_green_cells_attempt_09	amplitude	0	0.750	1	1	0

Campaign Best Run

- run: track1_forward_xgbm_ampl_h240_last_non_green_cells_attempt_08
- family: XGBM
- scope: amplitude
- harmonic: 240
- closure score: 0.750
- met / near / open: 1 / 1 / 0

Forward Benchmark Delta

	Surface	Before	After	Delta
Table 2	forward amplitude MAE	93G / 4Y / 3R	93G / 5Y / 2R	1Y / -1R
Table 3	forward amplitude RMSE	97G / 2Y / 1R	98G / 1Y / 1R	-1Y / 0R
Table 4	forward phase MAE	90G / 0Y / 0R	90G / 0Y / 0R	0Y / 0R
Table 5	forward phase RMSE	90G / 0Y / 0R	90G / 0Y / 0R	0Y / 0R

Canonical Forward Status After Closeout

- Table 2 : 5 yellow, 2 red
- Table 3 : 1 yellow, 1 red
- Table 4 : 0 yellow, 0 red
- Table 5 : 0 yellow, 0 red

Reference Archive Refresh

Direction	Family	Archived Targets	Unique Source Runs	Unique Source Configs	Archive Root
forward	SVM	19	11	11	models/paper_reference/rcim_track1/forward/svm_reference_models
forward	MLP	19	13	13	models/paper_reference/rcim_track1/forward/mlp_reference_models
forward	RF	19	12	12	models/paper_reference/rcim_track1/forward/rf_reference_models
forward	DT	19	12	12	models/paper_reference/rcim_track1/forward/dt_reference_models
forward	ET	19	12	12	models/paper_reference/rcim_track1/forward/et_reference_models
forward	ERT	19	11	11	models/paper_reference/rcim_track1/forward/ert_reference_models
forward	GBM	19	11	11	models/paper_reference/rcim_track1/forward/gbm_reference_models
forward	HGBM	19	11	11	models/paper_reference/rcim_track1/forward/hgbm_reference_models
forward	XGBM	19	12	12	models/paper_reference/rcim_track1/forward/xgbm_reference_models
forward	LGBM	19	17	17	models/paper_reference/rcim_track1/forward/lgbm_reference_models

Linked Artifacts

- benchmark report: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- training results master summary: `doc/reports/analysis/Training Results Master Summary.md`
- track1 reference root: `models/paper_reference/rcim_track1/forward/`
- campaign leaderboard:
`output/training_campaigns/track1/exact_paper/forward_last_non_green_cells/track1_forward_last_non_green_cells_campaign_2026-04-28_11_36_05/campaign_leaderboard.yaml`
- campaign best run YAML:
`output/training_campaigns/track1/exact_paper/forward_last_non_green_cells/track1_forward_last_non_green_cells_campaign_2026-04-28_11_36_05/campaign_best_run.yaml`
- campaign best run Markdown:
`output/training_campaigns/track1/exact_paper/forward_last_non_green_cells/track1_forward_last_non_green_cells_campaign_2026-04-28_11_36_05/campaign_best_run.md`